

# ASSIGNMENT: SWITCHING AND VLANs

*Network Configuration, VLAN Segmentation, and Automation*

This assignment covers practical aspects of Layer 2 switching, Virtual Local Area Networks (VLANs), port membership configuration, inter-VLAN connectivity isolation, and switch port range automation using Cisco IOS command-line interface syntax.

## Task 1: Network Diagram & Device VLAN Segmentation

Draw or layout a simple network topology showing three distinct end-devices connected to a central managed switch. Each device is assigned to a specific VLAN to isolate broadcast domains and enhance security.

| Device Name | Device Type        | Assigned VLAN | VLAN Name / Purpose |
|-------------|--------------------|---------------|---------------------|
| Laptop      | Personal Computer  | VLAN 20       | Personal Devices    |
| Smart Phone | Mobile Device      | VLAN 20       | Personal Devices    |
| Smart TV    | Media / IoT Device | VLAN 10       | Streaming & IoT     |

## Task 2: VLAN Creation and Inter-VLAN Isolation Verification

Configure a simulated Cisco switch (or NetSim/Packet Tracer equivalent) by creating VLAN 10 and VLAN 20, assigning respective ports to each VLAN, and verifying that hosts in different VLANs cannot communicate via direct Layer 2 forwarding.

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name STREAMING
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name PERSONAL
Switch(config-vlan)# exit
!
Switch(config)# interface range FastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
!
```

```
Switch(config)# interface range FastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end
Switch# write memory
```

**Verification Test:** From PC1 (VLAN 10, IP 192.168.10.10), pinging PC4 (VLAN 20, IP 192.168.20.20) will result in 'Request timed out' because traffic between different VLANs requires a Layer 3 routing device (Router-on-a-Stick or Multilayer Switch).

### Task 3: Moving a Port Between VLANs (Before & After)

Demonstrate moving PC3 from VLAN 10 (Port FastEthernet 0/3) to VLAN 20. This exercise illustrates how dynamic or static re-provisioning alters network reachability without physical recabling.

| Before Migration (VLAN 10)                                                                                                                                                                                                 | After Migration (VLAN 20)                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>Switch# configure terminal Switch(config)# interface FastEthernet 0/3 Switch(config-if)# switchport access vlan 10 Switch(config-if)# end</pre> <p>Result: PC3 can communicate with PC1 and PC2 (all in VLAN 10).</p> | <pre>Switch# configure terminal Switch(config)# interface FastEthernet 0/3 Switch(config-if)# switchport access vlan 20 Switch(config-if)# end</pre> <p>Result: PC3 can now communicate with PC4 and PC5 (VLAN 20). Pings to VLAN 10 devices fail.</p> |

### Task 4: Automating Port Assignment for VLAN 30

Write a script or Cisco IOS command sequence using range selection to efficiently assign 10 switch ports (FastEthernet 0/11 through 0/20) to VLAN 30 in a single operation, avoiding repetitive configuration steps.

```
! --- Cisco IOS Automation Script / Command List ---
enable
configure terminal
!
! 1. Create VLAN 30 and assign descriptive name
vlan 30
    name LAB_DEVICES
exit
!
! 2. Configure 10 ports simultaneously using interface range command
interface range FastEthernet 0/11 - 20
```

```
description Lab Workstation Ports
switchport mode access
switchport access vlan 30
spanning-tree portfast
exit
!
! 3. Save configuration
end
write memory
```